

148)

ξ	-2	-1	0	1	2
p	0,1	0,2	0,3	0,2	0,2

ξ^4	0	1	16
p	0,3	0,4	0,3

$$E\xi^4 = 0 \cdot 0,3 + 0,4 \cdot 1 + 0,3 \cdot 16 = 5,2$$

149)

ξ	0	1	2
p	0,4	0,2	0,4

η	-2	0	2
p	0,3	0,4	0,3

$$E(2\xi - \eta^2) = 2E\xi - E\eta^2 = -0,4$$

η^2	0	4
p	0,6	0,4

151) $\xi \in U(0;1)$ $\eta = a\xi + b, a > 0$

$$f_\xi(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases} \quad f_\eta(x) = |h'(x)| \cdot f_\xi(h(x))$$

$$y(x) = ax + b$$

$$\frac{y(x) - b}{a} = x$$

$$h(x) = \frac{x - b}{a}$$

$$h'(x) = \frac{1}{a}$$

$$f_\eta(x) = \begin{cases} 0, & x < b \\ \frac{1}{a}, & b \leq x \leq a+b \\ 0, & x > a+b \end{cases} \quad \eta = a\xi + b \in U(b; a+b)$$

152) $\xi \in E_\alpha$ $\eta = \beta\xi: \beta > 0$

$$f_\xi(x) = \begin{cases} 0, & x < 0 \\ e^{-\alpha x}, & x \geq 0 \end{cases} \quad f_\eta(x) = \frac{1}{\beta} f_\xi\left(\frac{x}{\beta}\right) \quad f_\eta(x) = \frac{1}{|\alpha|} f_\xi\left(\frac{x-a}{c}\right)$$

$$f_\eta(x) = \begin{cases} 0, & x < 0 \\ \frac{\alpha}{\beta} e^{-\frac{\alpha}{\beta}x}, & x \geq 0 \end{cases} \Rightarrow \eta \in E_{\frac{\alpha}{\beta}}$$

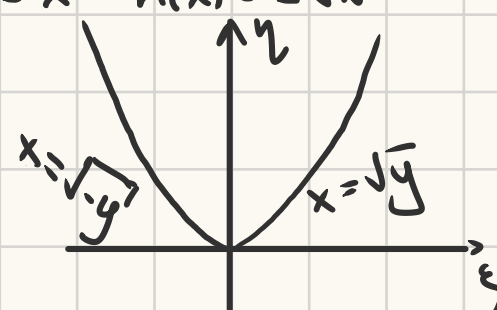
153) $\xi \in U(0,b)$ $\eta = \sqrt{\xi}$

$$f_\xi(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{b}, & 0 \leq x \leq b \\ 0, & x > b \end{cases}$$

$$f_\eta(x) = |h'(x)| \cdot f_\xi(h(x)) = \begin{cases} 0, & x < 0 \\ \frac{2x}{b}, & 0 \leq x \leq \sqrt{b} \\ 0, & x > \sqrt{b} \end{cases}$$

$y = \sqrt{x}$ $h(x) = x^2$ $h'(x) = 2x$

155) $\xi \in N(0,1)$ $\eta = \xi^2$ $g(x) = x^2$ $h(x) = \pm\sqrt{x}$ $h(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ \sqrt{-x}, & x < 0 \end{cases}$

$h_1(x) = \frac{1}{2\sqrt{x}}$ $f_\xi(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$  $\xi \in (-\infty; +\infty)$
 $h_2(x) = \frac{-1}{2\sqrt{x}}$ $\eta \in (0; +\infty)$

$$f_\eta(x) = \begin{cases} 0, & x \leq 0 \\ \frac{1}{2\sqrt{x}} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{(\sqrt{x})^2}{2}} + \frac{1}{2\sqrt{x}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(-\sqrt{x})^2}{2}}, & x > 0 \end{cases}$$

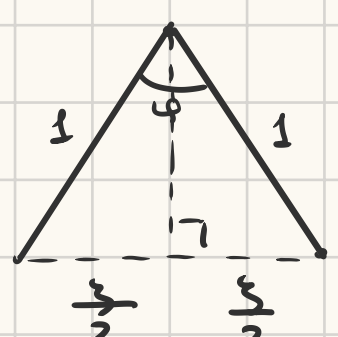
$$f_\eta(x) = \begin{cases} 0, & x \leq 0 \\ \frac{1}{\sqrt{2\pi x}} e^{-\frac{x}{2}}, & x > 0 \end{cases}$$

157) $\xi \in N(0,9)$ $\eta = (\xi - 2)^2$ $y(x) = (x-2)^2$ $\pm\sqrt{y(x)} = x-2 \pm\sqrt{y(x)+2} = x$
 $h(x) = \pm\sqrt{x} + 2$ $h'(x) = \pm\frac{1}{2\sqrt{x}}$

$$f_\eta(x) = |h'(x)| \cdot f_\xi(h(x))$$

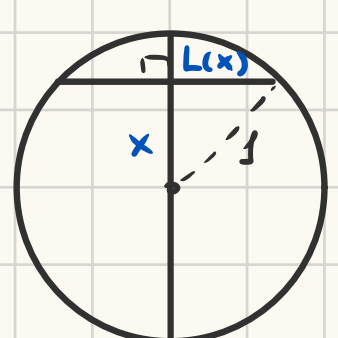
$$f_\xi(x) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{x^2}{18}}, x \in \mathbb{R}$$

$$f_\eta(x) = \frac{1}{2\sqrt{x} \cdot 3 \cdot \sqrt{2\pi}} e^{-\frac{(2+\sqrt{x})^2}{18}} = \frac{1}{6\sqrt{2\pi x}} e^{-\frac{(2+\sqrt{x})^2}{18}}$$

158)  $\varphi \in [0, \pi]$ $f_\varphi(x) = \begin{cases} 0, & \varphi < 0 \\ \frac{1}{\pi}, & 0 \leq \varphi \leq \pi \\ 0, & \varphi > \pi \end{cases}$ $\frac{\xi}{2} = \sin \frac{\varphi}{2}$
 $\xi = 2 \sin \frac{\varphi}{2}$
 $E g(\xi) = \int_{-\infty}^{+\infty} g(x) f_\xi(x) dx \Leftrightarrow$

$$\Leftrightarrow \int_0^{\frac{\pi}{2}} 2 \sin \frac{\varphi}{2} \cdot \frac{1}{\pi} d\varphi = \frac{4}{\pi} \cdot \frac{1}{2} \left(-\cos \frac{\varphi}{2} \right) \Big|_0^{\pi} = \frac{4}{\pi}$$

Омбер: $\frac{4}{\pi}$

159)  $\left(\frac{L(x)}{2}\right)^2 + x^2 = 1$ $L(x) = 2\sqrt{1-x^2}$ $L(x) = \int_{-1}^1 \frac{1}{2} \cdot 2\sqrt{1-x^2} dx = \frac{\pi}{2}$

$$f(x) = \begin{cases} 0, & x < -1 \\ \frac{1}{2}, & x \in [-1, 1] \\ 0, & x > 1 \end{cases}$$

Омбер: $\frac{\pi}{2}$